

NEWTON-RAPHSON METHOD

(i)

Let x_0 be an approximate root of $f(x) = 0$.

and let $x_1 = x_0 + h$ be the correct root so that

$$f(x_1) = f(x_0 + h) = 0.$$

$$\Rightarrow f(x_0 + h) = f(x_0) + h f'(x_0) + \frac{h^2}{2!} f''(x_0) + \dots$$

(Expansion by Taylor's series)

Neglecting second and higher order derivatives, we get

$$0 = f(x_0 + h) = f(x_0) + h f'(x_0).$$

$$\Rightarrow h = \frac{-f(x_0)}{f'(x_0)}.$$

Thus, we get

$$x_1 = x_0 + h = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$\text{Similarly ; } x_2 = x_1 + h = x_1 - \frac{f(x_1)}{f'(x_1)} \\ = x_0 + 2h$$

$$f(x_2) = f(x_1) + h f'(x_1) = 0$$

In general,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, \quad n = 0, 1, 2, \dots$$

Note : Newton-Raphson method has a second order of convergence.

Ex Use Newton-Raphson method to compute

(1) $(30)^{-1/5}$

(2) $\sqrt[3]{24}$

Solution (1.) let $x = (30)^{-1/5} \Rightarrow x^5 = \frac{1}{30}$

$\therefore f(x) = 30x^5 - 1 = 0$

Here, let us take $x_0 = \frac{1}{2} = 0.5$

$\therefore$ ~~$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$~~ $\Rightarrow x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

$= x_n - \frac{(30x_n^5 - 1)}{150x_n^4}$

$x_{n+1} = \frac{120x_n^5 + 1}{150x_n^4}$

$x_1 = \frac{120x_0^5 + 1}{150x_0^4} = \frac{120 \times \frac{1}{32} + 1}{150 \times \frac{1}{16}} = \frac{4.75}{9.375} = 0.5073$

$x_2 = \frac{120 \times 0.5073^5 + 1}{150 \times (0.5073)^4} = \frac{5.03186}{9.9346} = 0.5064$

(ii) let $x = \sqrt[3]{24}$

(3)

$$\Rightarrow x^3 - 24 = 0.$$

$\therefore$ let us suppose $x_0 = 3$.

$$x_{n+1} = x_n - \left(\frac{x_n^3 - 24}{3x_n^2} \right) = \frac{2x_n^3 + 24}{3x_n^2}.$$

$$x_1 = \frac{2 \times x_0^3 + 24}{3 \times x_0^2} = \frac{27 \times 2 + 24}{27} = \frac{78}{27} = 2.888.$$

$$x_2 = \frac{2 \times (2.888)^3 + 24}{3 \times (2.888)^2} = \frac{71.775}{24.8832} = 2.884$$

Ex Using Newton - Raphson method, derive formula to find $N^{1/q}$, $N > 0$, q is an integer, hence find

$18^{1/3}$ correct to four decimal places.

Solution

~~let~~ let $x = N^{1/q}$.

$$\Rightarrow x^q - N = 0.$$

$$\therefore x_{n+1} = x_n - \frac{x_n^q - N}{q x_n^{q-1}} = \frac{q x_n^q - x_n^q + N}{q x_n^{q-1}}$$

$$= \frac{(q-1)x_n^q + N}{q x_n^{q-1}}$$

Let us take $x_0 = 2.5$, $N = 18$, $q = 3$.

$$x_1 = \frac{(2)x_0^3 + 18}{3x_0^2} = \frac{2 \times (2.5)^3 + 18}{3 \times (2.5)^2} = \frac{49.25}{18.75} = 2.6266$$

$$x_2 = \frac{2 \times (2.6266)^3 + 18}{3 \times (2.6266)^2} = \frac{54.24197}{26.6970} = 2.62076$$

$$x_3 = \frac{2 \times (2.62076)^3 + 18}{3 \times (2.62076)^2} = \frac{54.0009}{20.6651} = 2.62075$$

Ex Using Newton-Raphson method obtain the real

root of the equation

(1) $x \sin x + \cos x = 0$

x_0

$\pi = 3.1416$

Ans

2.7984

(2) $3x - \cos x - 1 = 0$

1

0.607

(3) $x^3 - 5x + 3 = 0$

(0, 1)

$x_0 = 0.5$

0.65

$$x_{n+1} = x_n - \frac{x_n^3 - 5x_n + 3}{3x_n^2 - 5} = \frac{2x_n^3 - 3}{3x_n^2 - 5}$$

$$x_1 = \frac{2 \times (0.5)^3 - 3}{3 \times (0.5)^2 - 5} = \frac{-2.75}{-4.25} = 0.6470$$

$$x_2 = \frac{2 \times (0.6470)^3 - 3}{3 \times (0.6470)^2 - 5} = \frac{-2.4583}{-3.744133} \quad (4)$$

$$= 0.65656$$

$$x_3 = \frac{2 \times (0.65656)^3 - 3}{3 \times (0.65656)^2 - 5} = \frac{-2.43395}{-3.70678} = 0.65662$$

Ex . Determine the root of the equation $\cos x - x e^x = 0$ using Newton-Raphson method correct to four places of decimal.

Solution

$$x_{n+1} = x_n + \frac{\cos x_n - x_n e^{x_n}}{\sin x_n + (1+x_n)e^{x_n}}$$

$$x_1 = 0.65368$$

$$x_2 = 0.531343$$

$$x_3 = 0.51791$$

$$x_4 = 0.51776$$

ORDER OF Convergence

~~Newton's~~ method

$\Rightarrow$

$$E_n = x_n - \alpha$$

$$E_{n+1} = x_{n+1} - \alpha$$

Convergence (6)
Analysis

$$E_{n+1} = \frac{x_n - \frac{f(x_n)}{f'(x_n)} - \alpha}{f'(x_n)}$$

$$= E_n - \frac{f(E_n + \alpha)}{f'(E_n + \alpha)}$$

Using Taylor's expansion, we get

$$E_{n+1} = E_n - \left\{ \frac{f''(\alpha) + E_n f'(\alpha) + \frac{E_n^2}{2} f''(\alpha) \dots}{f'(\alpha) + E_n f''(\alpha) + \dots} \right\}$$

$$= E_n - E_n \left\{ \frac{f'(\alpha) + \frac{E_n}{2} f''(\alpha) \dots}{f'(\alpha) + E_n f''(\alpha) \dots} \right\}$$

$$= E_n - E_n \left\{ 1 + \frac{\frac{E_n}{2} f''(\alpha)}{f'(\alpha)} + \dots \right\}$$

$E_n - E_n \left(1 + \frac{E_n}{2} \frac{f''(\alpha)}{f'(\alpha)} \right)$

$$E_{n+1} = \frac{E_n^2 \frac{f''(\alpha)}{2 f'(\alpha)}}{1} \approx E_n^2 K$$

$$\therefore p = 2.$$

$\therefore$ NR method has second order convergence.